

Jongmin Sung, PhD

AI Research Scientist, Bio AI & Drug Discovery

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Professional Summary

Bio AI Scientist and ML Systems Architect with 15+ years of experience spanning computational drug discovery, single-molecule biophysics, production machine learning, protein modeling, and agentic AI. Holds a PhD in Applied Physics from Stanford and combines deep experimental biology expertise with scalable ML engineering and scientific agent development. Previously built production ML systems at Roche and currently leads a cardiac drug discovery collaboration with Stanford researchers, developing AI systems that integrate disease biology, protein structures, compound data, scientific literature, and functional assays. Focused on advancing drug discovery and protein design through biomedical foundation models and scientific AI agents.

Core Expertise

Bio AI and drug discovery | Agentic AI workflows | Genomics and precision healthcare | Biomedical foundation models | Protein modeling and protein language models | Scientific reasoning and LLM evaluation | Production ML and MLOps

Professional Experience

Stanford Cardiovascular Institute (CVI) — Lead AI Research Scientist, Stanford, CA *06/2026 – Present*

- Lead a cardiac drug discovery collaboration with Stanford cardiac researchers and cardiologists.
- Build scientific agents integrating disease biology, protein structures, compounds, literature, and functional assays.
- Develop protein modeling and ligand prediction pipelines for candidate prioritization and experimental validation.

Sailplane — Senior Knowledge Engineer, San Francisco, CA *02/2026 – 06/2026*

- Built structured knowledge systems for hierarchical AI agents, translating complex operational workflows into machine-readable representations for planning, validation, and decision support.

Addition Therapeutics — ML Protein Engineer (Consultant), South San Francisco, CA *07/2025 – 03/2026*

- Advanced PRINT genomic medicine platforms through AI-guided protein engineering for precise genome insertion.
- Built scalable, production-ready few-shot active learning ML pipelines using protein language models from scratch.

Aten Security — Applied AI Research Engineer (Consultant), San Francisco, CA *07/2025 – 01/2026*

- Designed and deployed LLM-powered systems for secure, real-time communication in highly trusted environments.
- Built AI pipelines integrating security-first design, safety constraints, backend services, and cloud infrastructure.

Scale AI — Human Frontier Collective Specialist, GenAI (Part-time) *03/2025 – Present*

- Evaluate frontier models on advanced scientific, coding, multimodal, and reasoning tasks.

Roche Sequencing Solutions — Principal Data Scientist, Santa Clara, CA *10/2019 – 05/2025*

- Built SBX sequencing platform, contributing to a Guinness World Record human genome sequencing in 4 hours.
- Applied biophysics, data science, and deep learning to improve next-generation sequencing performance.
- Developed sequencing software pipelines, driving gains in accuracy, read length, and throughput.

UCSF — Postdoctoral Research Fellow, San Francisco, CA (Advisor: Ron Vale) *06/2014 – 10/2019*

- Developed ML-driven analysis pipelines to model APC cell-cycle regulation mechanisms.
- Characterized human axonemal dynein motility through DNA origami-based in vitro reconstitution.

Stanford University — Research Assistant, Stanford, CA (Advisor: James Spudich) *04/2008 – 05/2014*

- Invented harmonic force spectroscopy to reveal the force-velocity relationship in muscle contraction.
- Conducted single-molecule studies on human β -cardiac myosin to investigate hypertrophic cardiomyopathy.

Education

- **PhD / MS in Applied Physics** | Stanford University, Stanford, CA
- **BS in Physics** | POSTECH, Pohang, South Korea

Technical Skills

- **Bio AI & Scientific Computing:** Drug discovery, Protein engineering, Genomics, Transcriptomics, AlphaFold, Boltz-2, PyMOL, BioPython, Statistical modeling, Bayesian inference, Monte Carlo simulation, Signal processing
- **AI & ML:** PyTorch, TensorFlow, scikit-learn, Transformers, LLMs, PLMs, ESM, GNNs, self-supervised learning, few-shot learning, MIL, prompt and context engineering
- **AI Engineering & MLOps:** Claude Code, Harness, agentic coding, hierarchical agent planning, LLM evaluation, GCP, AWS, Azure, Docker, Terraform, CI/CD
- **Programming & Data:** Python, NumPy, Pandas, Go, SQL, Git, Linux, Bash, backend development, API design

Selected Projects

- **JSAgent:** AI agent for hard mathematical optimization, evaluated against the Einstein Arena benchmark of unsolved problems in packing, autocorrelation, sphere geometry, and additive combinatorics. (2026) [GitHub]
- **Precision Health Agents:** Multi-agent systems for precision health. (2026) [GitHub]
- **Pancreatic Cancer Survival Prediction:** Predicted patient outcomes using self-supervised learning and multiple instance learning frameworks, integrating TCGA and external datasets. (2025) [GitHub]

Selected Advisory

- **Bio-AI / healthcare advisory:** Korean hospital AI strategy & clinical adoption (Valley Innovation Partners, 2026–present); early-stage Bio-AI & therapeutics scouting (NanoStar VC, 2026–present); US market & digital-therapeutics R&D for health-care startups (KITECH & KEIT, 2020–21).

Selected Publications

- **Sung, J.** (2026). *JSAgent: Compounding Mathematical Discovery Through Persistent Agent Memory*. arXiv (submitted).
- Hartooni, N., **Sung, J.**, Jain, A., & Morgan, D. O. (2022). Single-molecule analysis of specificity and multivalency in binding of short linear substrate motifs to the APC/C. *Nature Communications*, 13(1), 1–13.
- Niekamp, S., **Sung, J.**, Huynh, W., Bhabha, G., Vale, R., & Stuurman, N. (2019). Nanometer-accuracy distance measurements between fluorophores at the single-molecule level. *PNAS*, 116(10), 4275–4284.
- **Sung, J.**, et al. (2015). Harmonic force spectroscopy measures load-dependent kinetics of individual human β -cardiac myosin molecules. *Nature Communications*, 6(1), 7931.
- Mortensen, K. I.*, **Sung, J.***, Flyvbjerg, H., & Spudich, J. A. (2015). Optimized measurements of separations and angles between intra-molecular fluorescent markers. *Nature Communications*, 6(1), 8621. (* equal contributions)
- Sommese, R. F.*, **Sung, J.***, et al. (2013). Molecular consequences of the R453C hypertrophic cardiomyopathy mutation on human β -cardiac myosin motor function. *PNAS*, 110(31), 12607–12612. (* equal contributions)

Awards & Honors

- 1st place for innovation (among 45 teams from 21 countries), RocheHack Hackathon — 2020
- NRSA/F32 Postdoctoral Fellowship, National Institutes of Health — 2015–2018
- Bio-X Graduate Student Fellowship, Stanford University — 2009–2012